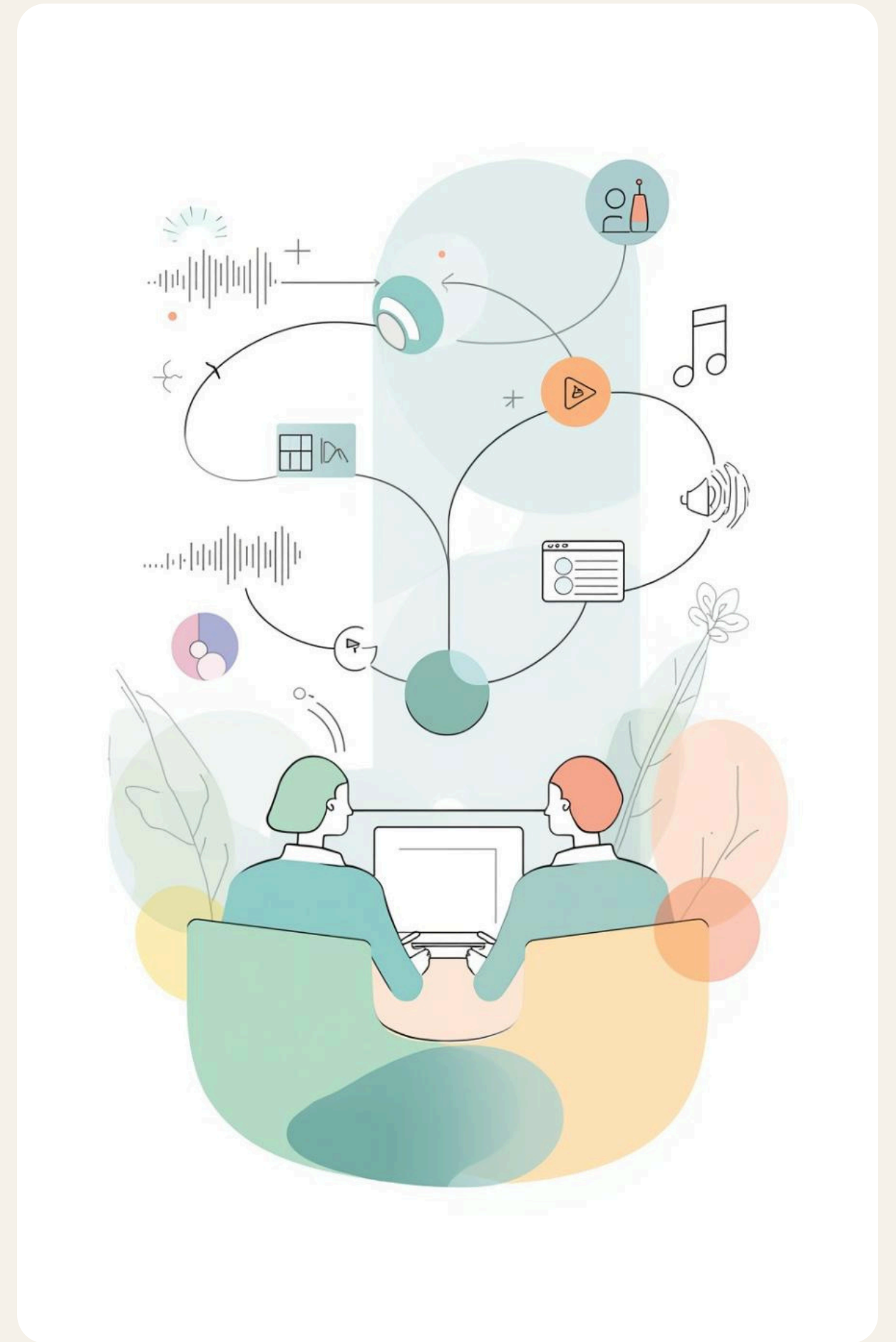


AI Lecture Assistant

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23FE10ITE00045



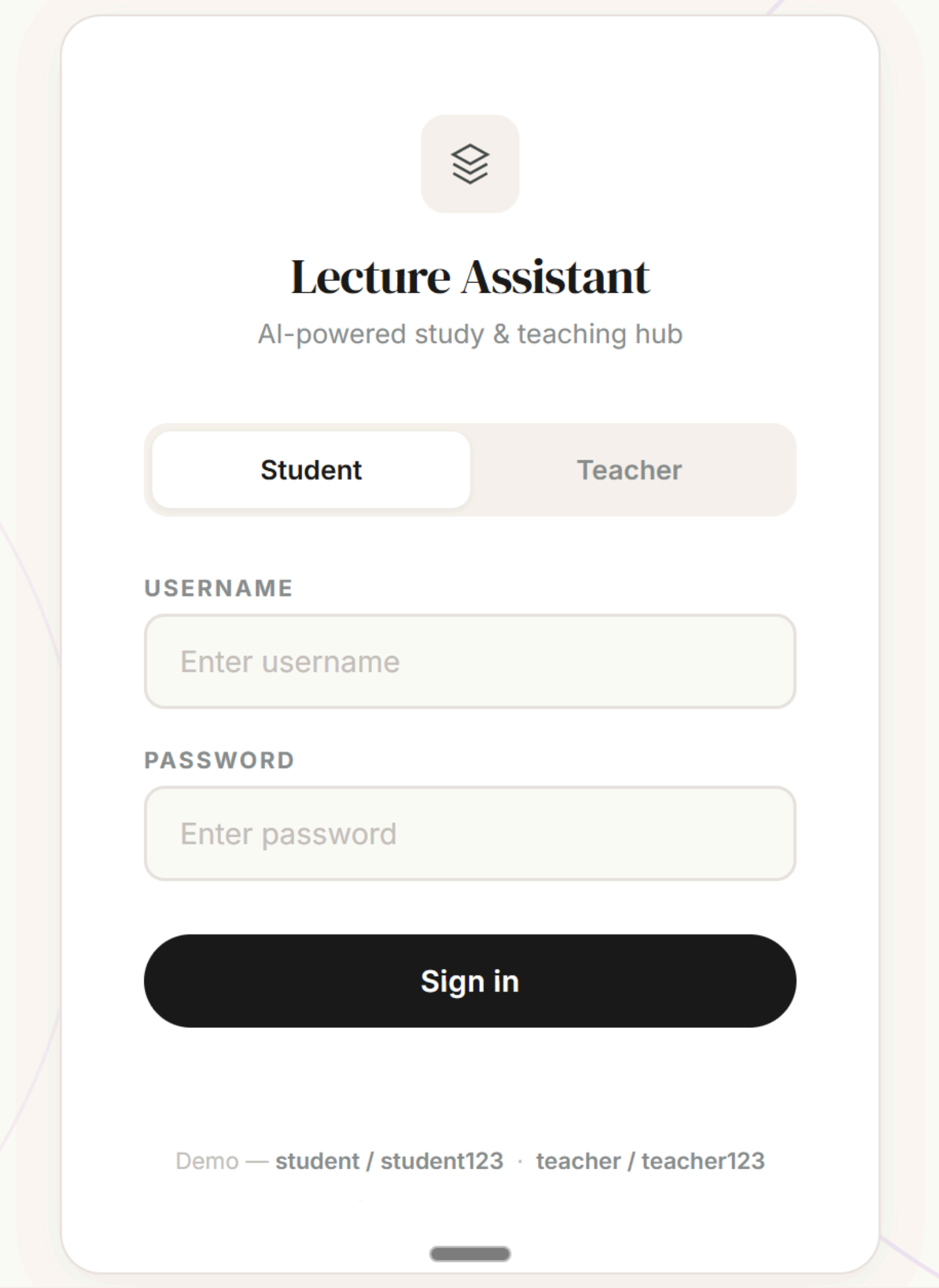
Addressing Challenges in Note-Taking

Students often struggle to capture key concepts during lectures, leading to gaps in comprehension. Our solution aims to enhance learning experiences and provide tools for both students and teachers to bridge this divide.



What is the AI Lecture Assistant?

The AI Lecture Assistant is a **web-based application** that transforms lecture recordings into comprehensive study materials, enhancing student comprehension and engagement through automated transcripts, notes, and quizzes.

A mobile application interface for 'Lecture Assistant'. At the top is a logo consisting of three stacked squares. Below the logo is the title 'Lecture Assistant' in a bold, black serif font, followed by the subtitle 'AI-powered study & teaching hub' in a smaller, grey sans-serif font. There are two toggle buttons: 'Student' (which is currently selected and highlighted with a white background and a thin grey border) and 'Teacher' (which has a grey background). Below these are two input fields: 'USERNAME' and 'PASSWORD', each with a light yellow background and a thin grey border. The 'USERNAME' field contains the placeholder text 'Enter username' and the 'PASSWORD' field contains 'Enter password'. At the bottom of the form is a large, black, rounded rectangular button with the text 'Sign in' in white. At the very bottom of the screen is a footer line of text: 'Demo — student / student123 · teacher / teacher123'. The entire interface is set against a light beige background with a subtle purple line on the left side.

How It Works: Steps 1-3

The AI Lecture Assistant begins with students uploading audio files or links. The system then transcribes the audio, generating structured notes and key concepts using advanced AI technology for effective learning.

Upload your lecture

Drop an audio file or paste a YouTube link to get started.

Upload File

Paste URL



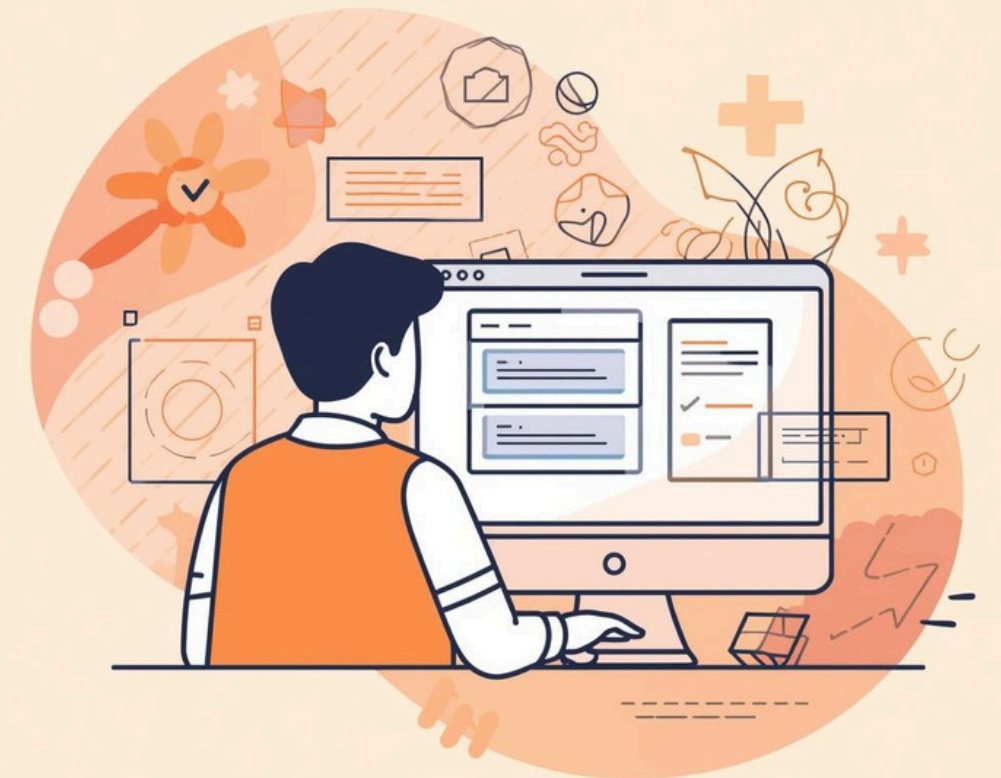
Drop audio file or click to browse

MP3, WAV, M4A — up to 100MB

Process Lecture

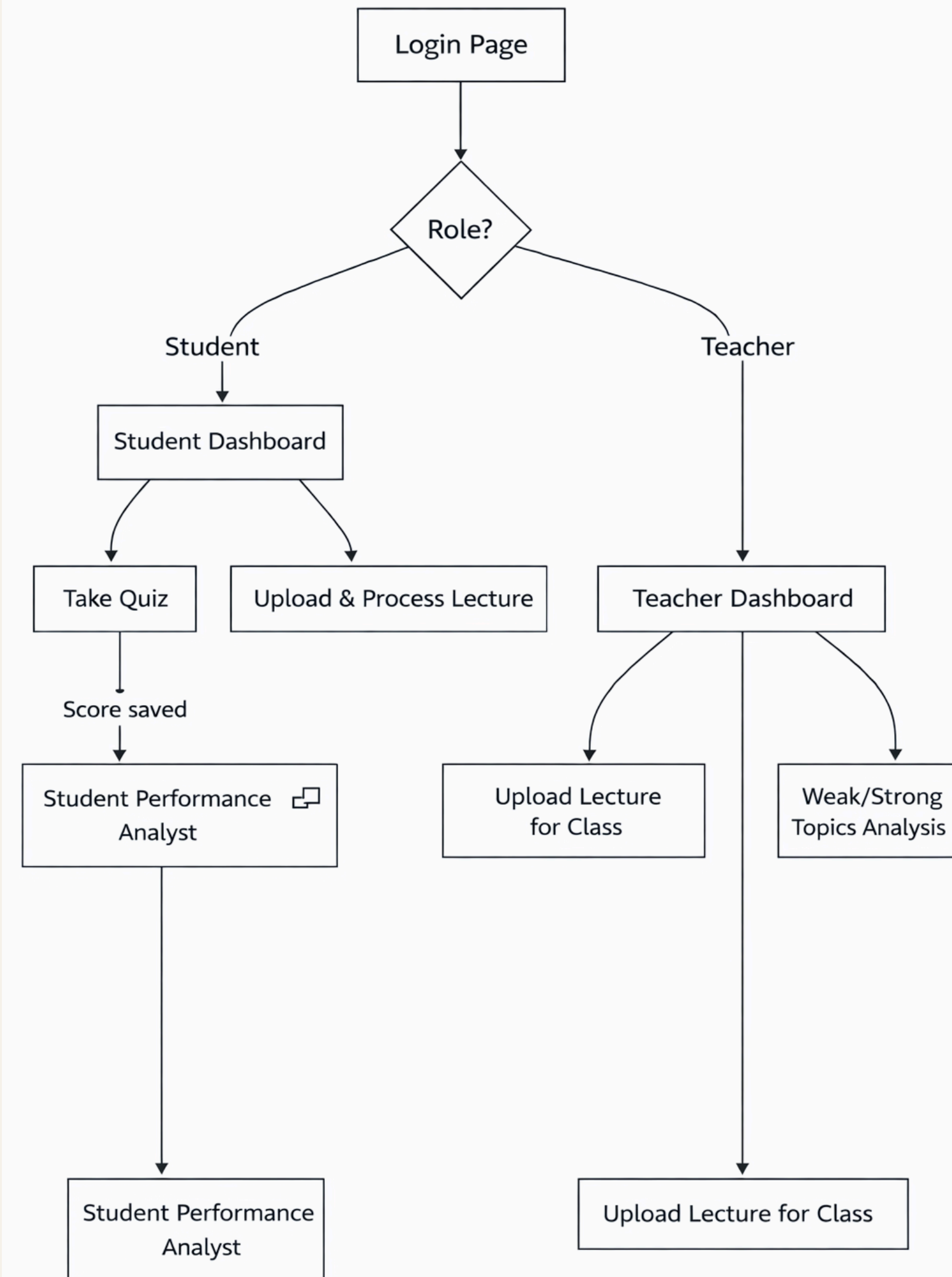
From Concepts to Quizzes

AI Givocaveer Key Concept extraction
from any audio Transcript



System Architecture Overview

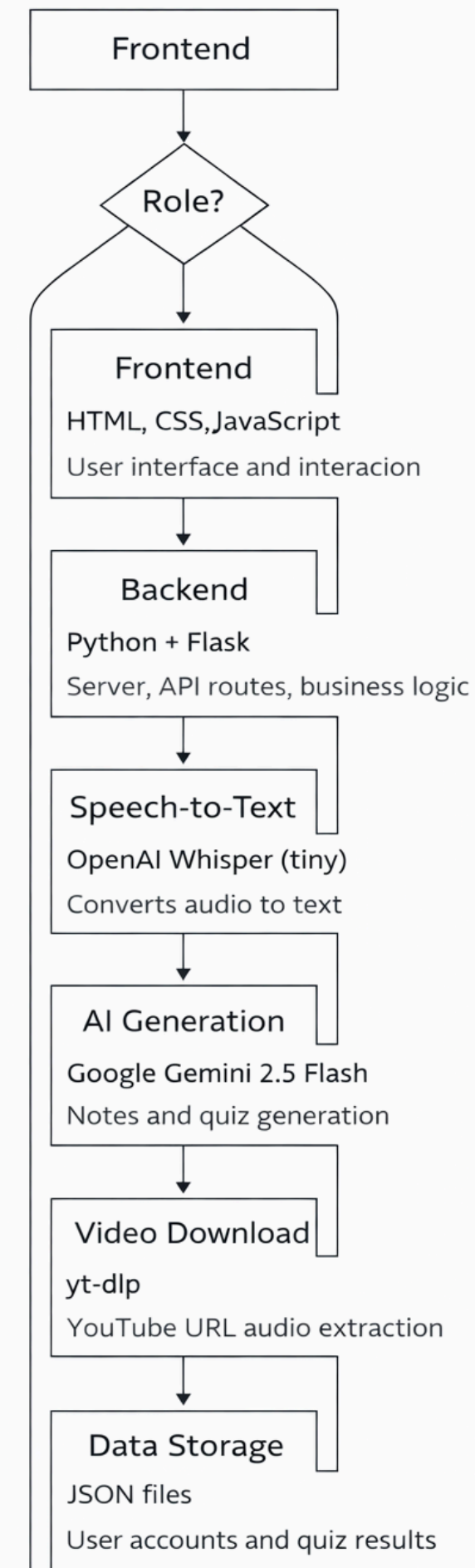
The AI Lecture Assistant integrates multiple AI technologies through a centralized system, employing Python Flask for the backend, ensuring seamless communication between transcription, note generation, and audio analysis components.



Technology Stack Overview

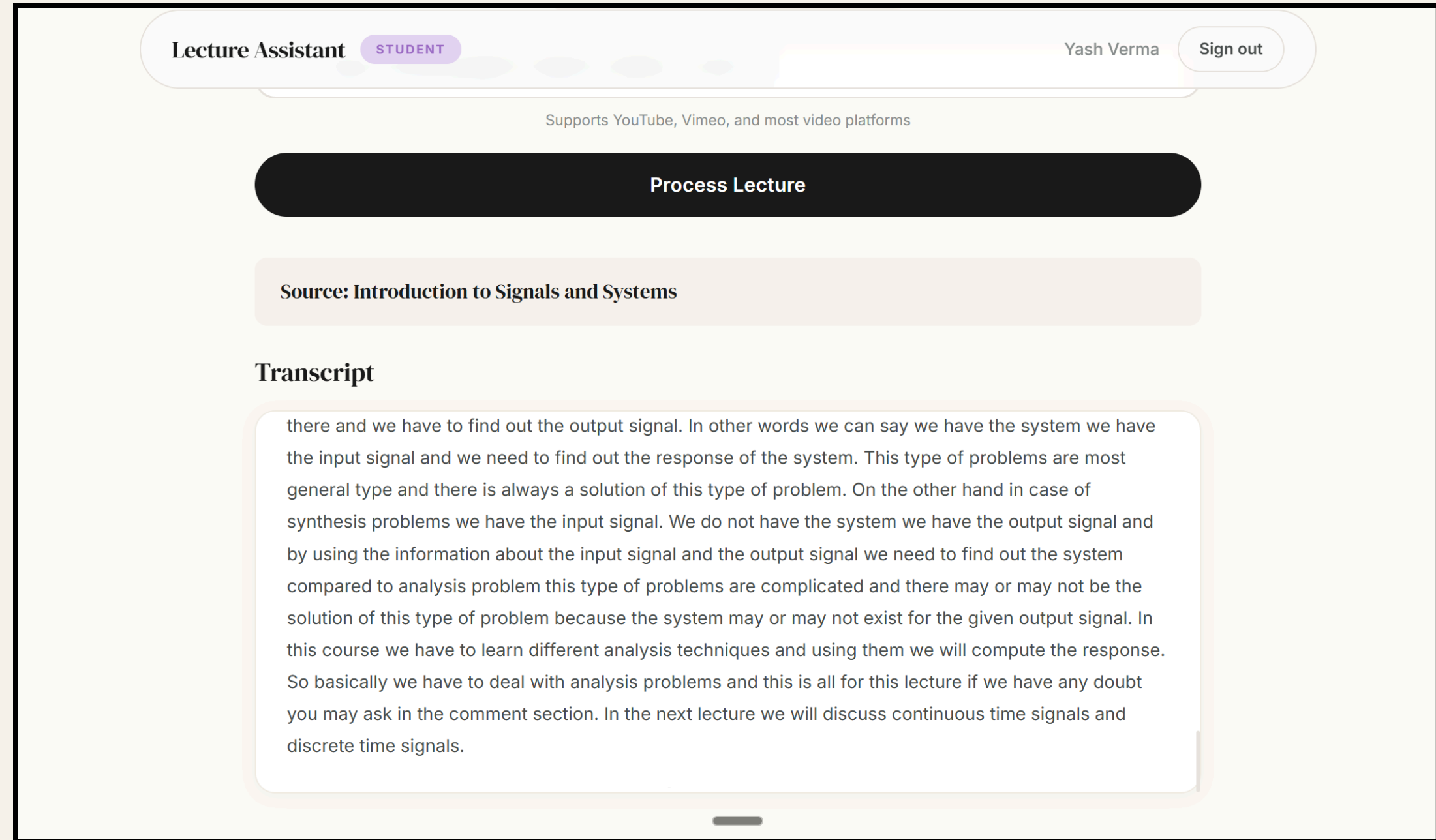
The AI Lecture Assistant is built using a robust technology stack, including Python Flask for backend development, HTML, CSS, and JavaScript for the frontend, and various AI tools for audio processing and analysis.

Technology Stack



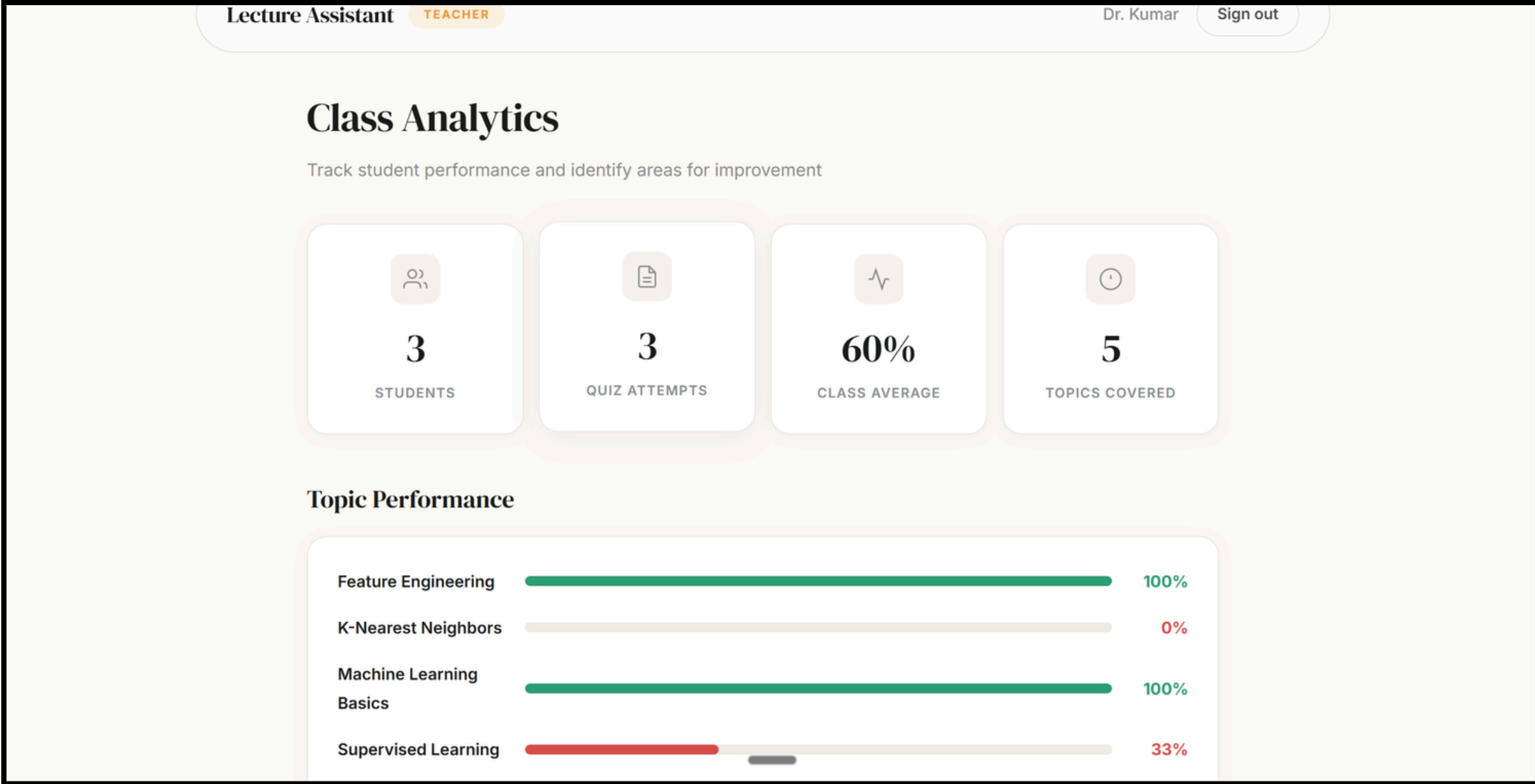
Student Dashboard: AI-Powered Learning

The student dashboard offers a seamless experience with features like **instant transcripts**, **structured notes**, and a timed quiz, empowering learners to engage with their lectures and improve comprehension effectively.



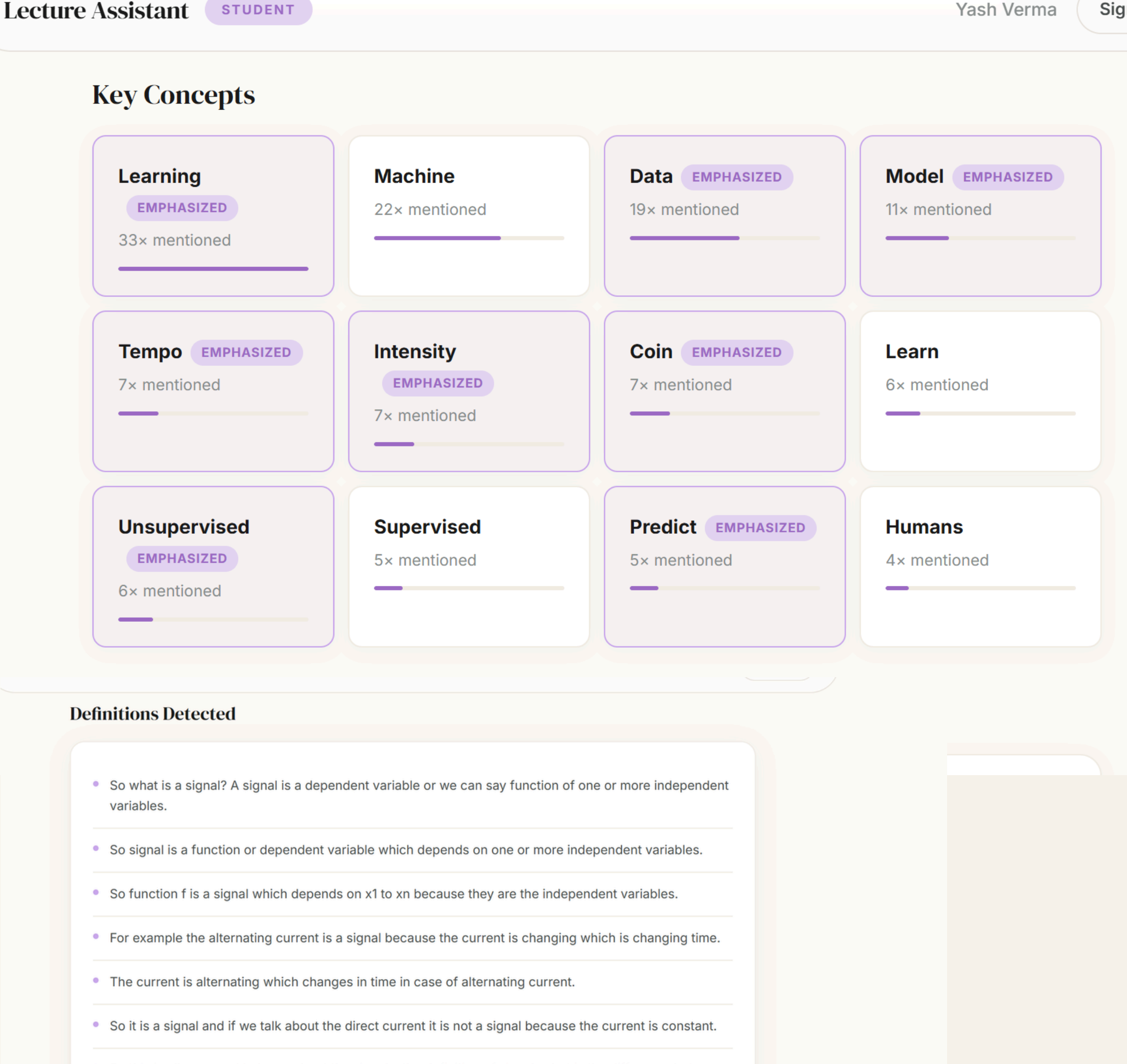
Teacher Dashboard Insights

The teacher dashboard offers professors a **comprehensive view** of student performance, visualizing summary statistics, topic accuracy, and individual results, ensuring efficient tracking of classroom engagement without manual grading.

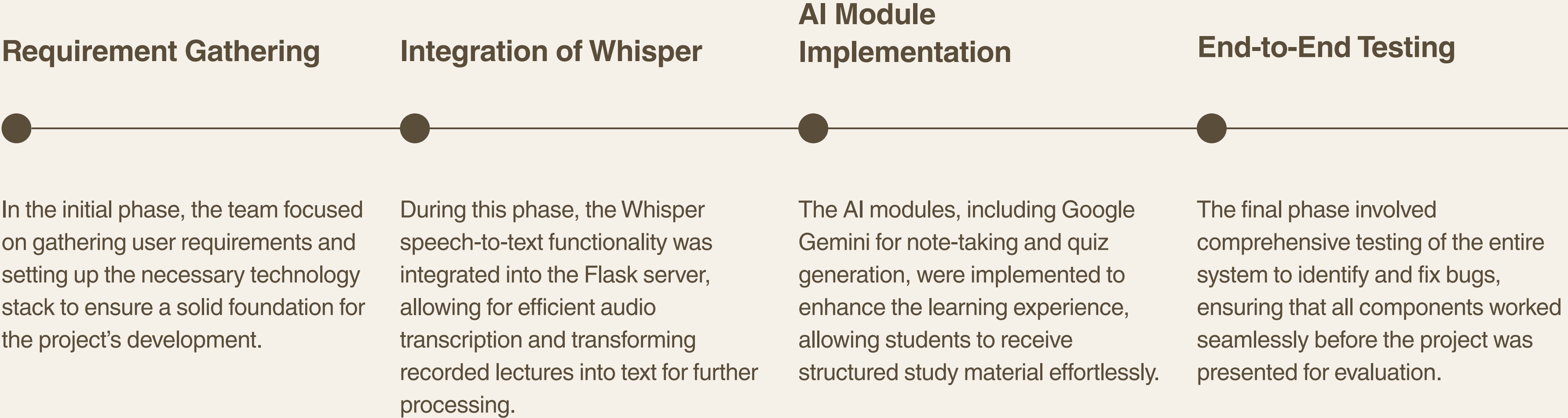


Key Features of Our Solution

Our AI Lecture Assistant delivers **ten essential features**, including audio uploads, AI transcription, structured notes, voice analysis, interactive quizzes, and insightful dashboards for both students and teachers, enhancing the learning experience.



Development Timeline



Future Focus

Embracing AI for educational
advancement

